

Coding & Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	AI-Powered Lead Generation System
Maximum Marks	5 Marks

Coding & Solution

The **AI-Powered Lead Generation System** is developed using **Python, Streamlit, Machine Learning, and Data Analytics** to identify, analyze, and prioritize potential customer leads. The system uses Machine Learning algorithms to predict lead quality, generate lead scores, and provide intelligent recommendations for improving sales conversion. The project includes data preprocessing, lead prediction, lead scoring, recommendation generation, and an interactive web-based interface. The code follows a modular architecture with reusable components, proper error handling, and clean coding practices to ensure reliability, scalability, and maintainability.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/hemamalinibora4-sketch/AI-Powered-Lead-Generation-System
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib

Key Features Implemented	AI-based lead prediction, lead scoring, customer data preprocessing, intelligent lead recommendations, interactive Streamlit web interface, real-time lead analysis
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Pending / Incomplete Features	User authentication, database integration, CRM integration, cloud deployment, email notifications, advanced analytics dashboard
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , activate the virtual environment (if applicable), run <code>streamlit run app.py</code> , and open the local Streamlit URL (typically <code>http://localhost:8501</code>)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository (GitHub)	Yes

Additional Notes / Comments:

The project follows a clean and modular coding structure with proper documentation, meaningful variable names, and error handling. It successfully analyzes customer data, predicts potential leads, generates lead scores, and provides intelligent recommendations for sales teams. The application is easy to maintain and can be enhanced with features such as user authentication, CRM integration, cloud deployment, advanced analytics, and automated email notifications in future versions.